

Tianxin Xie

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SKILLS

Languages: C++, Java, Python, JavaScript, HTML

Frameworks & Libraries: React, Java Swing, PyMuPDF, sentence-transformers, FAISS, Gemini 2.0 API

Developer Tools: Git, CMake, Visual Studio, VS Code, IntelliJ, Linux

PROJECTS

IEEE Document Knowledge Base  | Python, FAISS, Gemini 2.0 API, PyMuPDF Jan.2026

A Python intelligent research assistant that transforms complex academic papers into direct answers

- Used **PyMuPDF** as a tailored parser to ingest, clean, and chunk IEEE articles with double-column layouts.
- Built a vector database using **FAISS** and **Sentence-Transformers** for semantic retrieval of documents.
- Leveraged **Gemini 2.0 API** generate concise summaries of academic insights from retrieved contexts.

CalmCare System | Arduino C++, RP2040 Pico Dec. 2025

An Arduino-based patient monitoring and alert system to reduce hospital-induced delirium

- Integrated multiple sensors for multi-signal data acquisition, enabling continuous monitoring with **Arduino**.
- Implemented **UART** to transmit data to an Arduino for processing and threshold-based analysis.
- Interfaced an **RP2040 Pico** with an **LED display** to serve as a real-time user interface for intuitive data.

Laptop Store  | Java, Swing, OpenCSV, JFreeChart, HTML Feb. 2025

A Java Swing program that recommends suitable laptop models based on user requirements

- Collected and organized online computer data and integrated it into the Java application using **OpenCSV**.
- Developed an interactive UI leveraging **JFreeChart** to visualize and simplify hardware specs.
- Added **HTML** links for cross-platform price comparisons, enabling direct access to the lowest-priced options.

Super Pac-rio Bros Game  | Java, Swing, OOP Sep. 2024

A Java MCV structured mini-game inspired by Pac-Man and Mario, featuring maze navigation and coin collection.

- Implemented **object-oriented** design for characters, maze, and coins to enable modular and reusable code.
- Created a multi-input control system by integrating **KeyListener**, **MouseListener** and **javax.sound API**.
- Developed a local data persistence layer using **Java File I/O** to store and retrieve player game records.

EXPERIENCE

Java and C++ Programming Mentor Sep. 2024 - Jun. 2025

- Supported students in **Java OOP**, **Swing GUI** reinforcing design principles and coding best practices.
- Tutored C++ for NOIP, USACO, CCC, covering **DP**, **linked lists**, **OOP data structures**, and **pointers**.
- Supported high school female students in coding projects, **promoting women in computer science**.

Math Competition Club – Executive Jan. 2023 – Jun.2025

- Organized and scheduled weekly Math Club meetings to ensure members' consistent engagement.
- Demonstrated strong leadership by coordinating with members to plan club activities.

EDUCATION

University of Waterloo

Waterloo, ON

1B Bachelor of Applied Science in Computer Engineering

Sep. 2025 – Expected Apr. 2030